LABORATORY WORK #2: ADVANCED DDL OPERATIONS
Topic: Database Creation, Table Management & Data Types

PART 1: MULTIPLE DATABASE MANAGEMENT

TASK 1.1: DATABASE CREATION

```
CREATE DATABASE university_main
    WITH OWNER = CURRENT_USER
    TEMPLATE = template0
    ENCODING = 'UTF8';

CREATE DATABASE university_archive
    WITH TEMPLATE = template0
    CONNECTION LIMIT = 50;

CREATE DATABASE university_test
    WITH TEMPLATE = template0
    CONNECTION LIMIT = 10;

ALTER DATABASE university_test IS_TEMPLATE = TRUE;
```

TASK 1.2: TABLESPACE OPERATIONS

```
CREATE TABLESPACE student_data
    OWNER = CURRENT_USER
    LOCATION '/data/students';

CREATE TABLESPACE course_data
    OWNER = CURRENT_USER
    LOCATION '/data/courses';

CREATE DATABASE university_distributed
    WITH TEMPLATE = template0
    TABLESPACE = student_data
    ENCODING = 'LATIN9';
```

PART 2: COMPLEX TABLE CREATION
Connect to university_main before Part 2
TASK 2.1: UNIVERSITY MANAGEMENT SYSTEM

```
CREATE TABLE students (
    student_id SERIAL PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100),
    phone CHAR(15),
    date_of_birth DATE,
    enrollment_date DATE,
    gpa NUMERIC(4,2),
    is_active BOOLEAN,
    graduation_year SMALLINT
);

CREATE TABLE professors (
    professor_id SERIAL PRIMARY KEY,
    first_name VARCHAR(50),
    last_name VARCHAR(50),
    email VARCHAR(100),
    office_number VARCHAR(20),
    hire_date DATE,
    salary NUMERIC(12,2),
```

```
    is_tenured BOOLEAN,
    years_experience INTEGER
);

CREATE TABLE courses (
    course_id SERIAL PRIMARY KEY,
    course_code CHAR(8),
    course_title VARCHAR(100),
    description TEXT,
    credits SMALLINT,
    max_enrollment INTEGER,
    course_fee NUMERIC(10,2),
    is_online BOOLEAN,
    created_at TIMESTAMP WITHOUT TIME ZONE
);


TASK 2.2: TIME-BASED AND SPECIALIZED TABLES

CREATE TABLE class_schedule (
    schedule_id SERIAL PRIMARY KEY,
    course_id INTEGER,
    professor_id INTEGER,
    classroom VARCHAR(20),
    class_date DATE,
    start_time TIME WITHOUT TIME ZONE,
    end_time TIME WITHOUT TIME ZONE,
    duration INTERVAL
);

CREATE TABLE student_records (
    record_id SERIAL PRIMARY KEY,
    student_id INTEGER,
    course_id INTEGER,
    semester VARCHAR(20),
    year INTEGER,
    grade CHAR(2),
    attendance_percentage NUMERIC(4,1),
    submission_timestamp TIMESTAMP WITH TIME ZONE,
    last_updated TIMESTAMP WITH TIME ZONE
);


PART 3: ADVANCED ALTER TABLE OPERATIONS

TASK 3.1: MODIFY STUDENTS TABLE

ALTER TABLE students
    ADD COLUMN middle_name VARCHAR(30);

ALTER TABLE students
    ADD COLUMN student_status VARCHAR(20);

ALTER TABLE students
    ALTER COLUMN phone TYPE VARCHAR(20)
    USING RTRIM(phone);

ALTER TABLE students
    ALTER COLUMN student_status SET DEFAULT 'ACTIVE';

ALTER TABLE students
    ALTER COLUMN gpa SET DEFAULT 0.00;
```

```
MODIFY PROFESSORS TABLE

ALTER TABLE professors
    ADD COLUMN department_code CHAR(5);

ALTER TABLE professors
    ADD COLUMN research_area TEXT;

ALTER TABLE professors
    ALTER COLUMN years_experience TYPE SMALLINT;

ALTER TABLE professors
    ALTER COLUMN is_tenured SET DEFAULT FALSE;

ALTER TABLE professors
    ADD COLUMN last_promotion_date DATE;


MODIFY COURSES TABLE

ALTER TABLE courses
    ADD COLUMN prerequisite_course_id INTEGER;

ALTER TABLE courses
    ADD COLUMN difficulty_level SMALLINT;

ALTER TABLE courses
    ALTER COLUMN course_code TYPE VARCHAR(10)
    USING RTRIM(course_code);

ALTER TABLE courses
    ALTER COLUMN credits SET DEFAULT 3;

ALTER TABLE courses
    ADD COLUMN lab_required BOOLEAN DEFAULT FALSE;


TASK 3.2: MODIFY CLASS_SCHEDULE TABLE

ALTER TABLE class_schedule
    ADD COLUMN room_capacity INTEGER;

ALTER TABLE class_schedule
    DROP COLUMN duration;

ALTER TABLE class_schedule
    ADD COLUMN session_type VARCHAR(15);

ALTER TABLE class_schedule
    ALTER COLUMN classroom TYPE VARCHAR(30);

ALTER TABLE class_schedule
    ADD COLUMN equipment_needed TEXT;

MODIFY STUDENT_RECORDS TABLE

ALTER TABLE student_records
    ADD COLUMN extra_credit_points NUMERIC(4,1);

ALTER TABLE student_records
    ALTER COLUMN grade TYPE VARCHAR(5)
    USING RTRIM(grade);

ALTER TABLE student_records
```

```
    ALTER COLUMN extra_credit_points SET DEFAULT 0.0;

ALTER TABLE student_records
    ADD COLUMN final_exam_date DATE;

ALTER TABLE student_records
    DROP COLUMN last_updated;


PART 4: TABLE RELATIONSHIPS AND MANAGEMENT
TASK 4.1: CREATE DEPARTMENTS TABLE

CREATE TABLE departments (
    department_id SERIAL PRIMARY KEY,
    department_name VARCHAR(100),
    department_code CHAR(5),
    building VARCHAR(50),
    phone VARCHAR(15),
    budget NUMERIC(14,2),
    established_year INTEGER
);


CREATE LIBRARY_BOOKS TABLE

CREATE TABLE library_books (
    book_id SERIAL PRIMARY KEY,
    isbn CHAR(13),
    title VARCHAR(200),
    author VARCHAR(100),
    publisher VARCHAR(100),
    publication_date DATE,
    price NUMERIC(10,2),
    is_available BOOLEAN,
    acquisition_timestamp TIMESTAMP WITHOUT TIME ZONE
);

CREATE STUDENT_BOOK_LOANS TABLE

CREATE TABLE student_book_loans (
    loan_id SERIAL PRIMARY KEY,
    student_id INTEGER,
    book_id INTEGER,
    loan_date DATE,
    due_date DATE,
    return_date DATE,
    fine_amount NUMERIC(10,2),
    loan_status VARCHAR(20)
);


TASK 4.2: ADD INTEGRATION COLUMNS

ALTER TABLE professors
    ADD COLUMN department_id INTEGER;

ALTER TABLE students
    ADD COLUMN advisor_id INTEGER;

ALTER TABLE courses
    ADD COLUMN department_id INTEGER;

CREATE GRADE_SCALE TABLE
```

```
CREATE TABLE grade_scale (
    grade_id SERIAL PRIMARY KEY,
    letter_grade CHAR(2),
    min_percentage NUMERIC(4,1),
    max_percentage NUMERIC(4,1),
    gpa_points NUMERIC(3,2)
);

CREATE SEMESTER_CALENDAR TABLE

CREATE TABLE semester_calendar (
    semester_id SERIAL PRIMARY KEY,
    semester_name VARCHAR(20),
    academic_year INTEGER,
    start_date DATE,
    end_date DATE,
    registration_deadline TIMESTAMP WITH TIME ZONE,
    is_current BOOLEAN
);


PART 5: TABLE DELETION AND CLEANUP
TASK 5.1: DROP TABLES IF THEY EXIST

DROP TABLE IF EXISTS student_book_loans;
DROP TABLE IF EXISTS library_books;
DROP TABLE IF EXISTS grade_scale;


RECREATE GRADE_SCALE WITH DESCRIPTION

CREATE TABLE grade_scale (
    grade_id SERIAL PRIMARY KEY,
    letter_grade CHAR(2),
    min_percentage NUMERIC(4,1),
    max_percentage NUMERIC(4,1),
    gpa_points NUMERIC(3,2),
    description TEXT
);


DROP SEMESTER_CALENDAR WITH CASCADE

DROP TABLE IF EXISTS semester_calendar CASCADE;

RECREATE SEMESTER_CALENDAR

CREATE TABLE semester_calendar (
    semester_id SERIAL PRIMARY KEY,
    semester_name VARCHAR(20),
    academic_year INTEGER,
    start_date DATE,
    end_date DATE,
    registration_deadline TIMESTAMP WITH TIME ZONE,
    is_current BOOLEAN
);


TASK 5.2: DATABASE CLEANUP

DROP DATABASE IF EXISTS university_test;
DROP DATABASE IF EXISTS university_distributed;

CREATE DATABASE university_backup
```

```
    WITH TEMPLATE = university_main;

END OF LABORATORY WORK #2
```